

MOBILE SHOP ERP — SAAS ARCHITECTURE PLAN

Multi-Tenant SaaS Transformation, KYC Verification & Super Admin Specifications

1. ARCHITECTURAL STRATEGY & GOAL

The objective of this project is to transform the single-tenant Mobile Shop ERP into a scalable, high-performance Multi-Tenant SaaS platform. This enables selling and onboarding multiple independent mobile shop owners seamlessly on a unified infrastructure with total data isolation.

2. SUPER ADMIN TENANT CONTROLS & MANAGEMENT

A. Pause / Suspend Shop Account

Instantly lock any shop owner and their staff from accessing the system (e.g. for overdue subscription payments). API requests return HTTP 403 Account Suspended.

B. Activate & Restore Account Access

Re-enable access for suspended shop accounts in 1-click upon payment confirmation or policy compliance.

C. Soft Delete Shop Account

Deactivate a shop account permanently while retaining historical financial and transaction logs for platform compliance.

D. Manual Owner Onboarding & Mandatory OTP Validation

Super Admin can manually create shop owner accounts directly from the SaaS portal and complete mandatory 6-digit email OTP verification to validate and activate the owner account.

E. Tenant Support Impersonation ("Login as Tenant")

Super Admin can generate a temporary secure session token to view any shop owner workspace for customer support and troubleshooting.

3. SHOP KYC & BUSINESS DOCUMENT VERIFICATION SYSTEM

Owner Personal Verification

National ID (NID Number + Front & Back image upload), Passport-size photo, Secondary contact phone & relation.

Business & Legal Verification

Trade License Number, Issue/Expiration dates + Scanned document upload, TIN/BIN certificate number & upload.

Super Admin KYC Review Vault

Super Admin reviews submitted NID & Trade License inside DocumentVaultModal.jsx. Super Admin can Approve KYC or Reject with reasons.

4. PROPOSED SYSTEM COMPONENT CHANGES

A. Database Models (server/modules/tenant/)

- [NEW] tenant.model.js: Stores shop accounts, plan tiers, status (ACTIVE, PAUSED, PENDING_KYC, DELETED), kycDocuments (nidFront, nidBack, tradeLicense, tinCert), expiry date, branch & user limits.
- [NEW] subscription.model.js: Billing cycles, pricing tiers, and plan feature flags.
- [MODIFY] Inject tenantId: ObjectId across all 27+ ERP collections (Product, Sale, Purchase, Expense, Customer, User, etc.) for data isolation.

B. Backend API Middleware & Controllers

- [NEW] tenant.middleware.js: Automatically extracts tenantId from JWT / domain and scopes every MongoDB query filter.
- [NEW] tenant.controller.js: Endpoints for createTenant, pauseTenant, activateTenant, deleteTenant, verifyKyc, and impersonateTenant.
- [MODIFY] auth.service.js: Embeds tenantId inside JWT access tokens and refresh tokens.

C. Super Admin Control Portal (client/src/pages/SaaS/)

- [NEW] TenantManagement.jsx: Master dashboard to pause, activate, delete, validate, or onboard new shop owners.
- [NEW] DocumentVaultModal.jsx: Inspection modal for Super Admin to review NID, Trade License, and TIN documents.
- [NEW] SubscriptionPlans.jsx: Configure SaaS pricing tiers, branch limits, and feature access.

D. Self-Service Shop Onboarding (client/src/pages/Auth/)

- [NEW] RegisterShop.jsx: Public self-registration portal with multi-step wizard: 1. Shop Info, 2. NID Upload, 3. Trade License Upload, 4. Email OTP.

5. VERIFICATION & TESTING PLAN

1. Shop Registration & KYC Upload: Register a new shop owner with NID & Trade License uploads.
 2. Super Admin KYC Review: Log into SaaS portal, open Document Vault Modal, inspect uploaded NID and Trade License, and Approve KYC.
 3. Pause & Resume Test: Super Admin pauses Shop A. Verify Shop A users get HTTP 403 Account Suspended on login. Resume access and verify access is restored.
 4. Data Isolation Test: Log into Shop A and Shop B; verify zero data leakage.
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